
AttenA+: Velocity Field Action Attention for Enhancing Action-Centric Robotic Foundation Models

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Abstract

1 Action-centric robotic foundation models, including vision-language-action (VLA)
2 and world-action models (WAM), enable end-to-end robotic manipulation from
3 visual observations and language instructions. However, existing frameworks treat
4 all action timesteps equally during training, overlooking the inherent physical
5 hierarchy: slow movements correspond to critical, precision-demanding steps,
6 while fast movements are transitional and error-tolerant. This misalignment limits
7 performance on high-precision and long-horizon tasks. To address this issue, we
8 propose **AttenA+**, a generic training framework empowered by velocity field-
9 based action attention to prioritize slow and critical action segments. By assigning
10 higher learning weights to low-velocity actions, AttenA+ aligns training with real-
11 world manipulation physics and can be seamlessly plugged into existing models
12 without architectural modifications. Experiments show that AttenA+ achieves
13 state-of-the-art performance on Libero and RoboTwin benchmarks (98.6% and
14 93.1%), outperforming OpenVLA-OFT, $\pi_{0.5}$, and FastWAM. Real-world validation
15 on a Franka manipulator further confirms its effectiveness and generalization.
16 Beyond performance gains, AttenA+ provides a novel, cost-effective paradigm
17 complementary to data and model scaling by exploiting intrinsic structural priors in
18 action sequences. Our work highlights that attention should extend beyond inputs
19 to output action trajectories, inspiring future physics-aware action learning for
20 general robotic manipulation.

21 **Keywords:** Robotic Foundation Model; Action-Centric Learning; Velocity Field
22 Attention; Robotic Manipulation

1 Introduction

24 Action-centric robotic foundation models, particularly Vision-Language-Action (VLA) and World-
25 Action Models (WAM), have become the cornerstone of modern robotic control, enabling robots to
26 interpret multimodal instructions and execute complex physical manipulation tasks [3, 4, 5]. However,
27 two fundamental limitations hinder the further advancement of current paradigms:

28 First, the dominant route toward performance gains heavily relies on *scaling laws*—either expanding
29 the scale of training datasets [6, 7, 8, 9] or enlarging model capacity [3, 10, 5, 11, 12]. This strategy
30 directly inherits the scaling philosophy from natural language processing (NLP), where textual data
31 is treated as homogeneous, and performance improvements follow predictable power-law trends
32 with increasing data volume and parameter size [13, 14]. In standard NLP paradigms, all tokens
33 are assumed equally informative, and gains are driven purely by scale [15, 16]. In contrast, robotic
34 manipulation trajectories are inherently *heterogeneous*: not all action steps equally determine task
35 success, and blind scaling fails to exploit intrinsic structural regularities within action sequences
36 [17, 18]. Unlike linguistic tokens, robot action trajectories exhibit clear temporal and physical

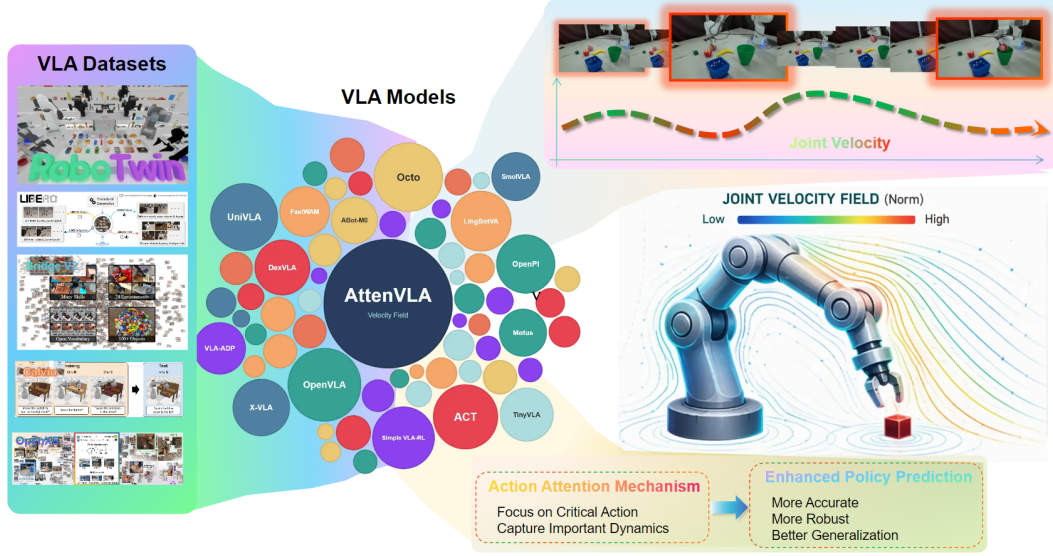


Figure 1: **Overview of AttenA+.** AttenA+ is a paradigm-agnostic enhancement framework for action-centric robotic foundation models, introducing velocity-field-based action attention to prioritize slow, critical manipulation steps. It seamlessly plugs into mainstream discriminative (e.g., OpenVLA-OFT) and generative (π_0 , $\pi_{0.5}$, Diffusion Policy) architectures, as well as emerging World-Action Models (WAM). Without modifying core backbones or relying on data/model scaling, AttenA+ generalizes across diverse robotic datasets including Libero [1] and RoboTwin [2], and consistently improves task success rates over state-of-the-art baselines.

37 importance hierarchies, rendering naive data and model scaling inefficient and misaligned with
38 real-world manipulation physics.

39 Second, existing action-centric foundation models confine attention mechanisms solely to input
40 modalities—such as visual features from RGB observations [3] and semantic embeddings from
41 language commands [4]—while treating output action trajectories as flat, equally weighted sequences.
42 This design deviates from human motor control principles: in practical manipulation, slow, deliberate
43 movements (e.g., precise gripper alignment with fragile objects, careful placement on constrained
44 surfaces) correspond to *task-critical key actions* that dominate success or failure. By comparison,
45 fast transitional motions (e.g., arm retargeting between waypoints) are relatively error-tolerant and
46 contribute less to final task outcomes. Ignoring this inherent action importance hierarchy leads to
47 suboptimal optimization: models waste representational capacity on trivial transitional motions while
48 under-fine-tuning the slow, precision-critical steps that truly govern manipulation performance.

49 Motivated by these observations, we argue that future gains for robotic foundation models should
50 go beyond blind scaling and toward principled *action sequence mining*: explicitly modeling the
51 varying importance of individual trajectory steps according to their physical properties. To this end,
52 we propose **AttenA+**, a general enhancement framework that introduces *velocity field-based action*
53 *attention* to reweight trajectory learning by motion speed. By assigning higher optimization priority
54 to low-velocity critical actions, AttenA+ aligns model training with both human motor intuition and
55 the physical nature of robotic manipulation.

56 Beyond empirical performance improvements, our work delivers a conceptual paradigm shift for
57 action-centric robotic foundation models. As illustrated in Figure 1, we advocate *action-centric*
58 *attention* as a complementary supplement to scaling laws: beyond pursuing larger datasets or bigger
59 backbones, we unlock the latent potential of existing trajectories by mining physical structural priors
60 within action sequences. Unlike NLP, where token importance stems primarily from semantics,
61 robot action importance is intrinsically rooted in physical dynamics such as velocity and precision
62 demand. AttenA+ validates that such physical action priors can yield consistent performance gains
63 without additional data collection or model scaling. This perspective opens new research avenues:

64 future designs can extend velocity-driven action attention to richer physical signals (e.g., force
 65 feedback, contact states, and jerk constraints), paving the way toward more efficient, physics-aware,
 66 and generalizable robotic manipulation systems.

67 2 Related Works

68 2.1 Action-Centric Robotic Foundation Models

69 Vision-Language-Action (VLA) and World-Action Models (WAM) enable end-to-end robotic manip-
 70 ulation by grounding language in visual observations to generate continuous action sequences. A
 71 wide range of VLA frameworks have been proposed to advance robotic manipulation performance,
 72 including foundational models and their variants, as well as specialized optimizations. OpenVLA [3]
 73 serves as a core foundational framework unifying visual perception, language understanding, and
 74 action generation, with its variant OpenVLA-OFT [19] further optimizing via orthogonal fine-tuning
 75 to push state-of-the-art (SOTA) performance on Libero tasks. The π model series, including π_0
 76 [5], $\pi_0 + \text{FAST}$ [20], and $\pi_{0.5}$ [12], advances generative VLA capabilities through flow matching
 77 for strong generalization. Other representative VLA models and optimizations include UniVLA
 78 [10], VLA-ADP [21], CogACT [22], SmolVLA [23], NORA and NORA-Long [24], WorldVLA
 79 and WorldVLA* [11], SP-VLA [25], FlashVLA [26], VLA-Cache [27], FastV and FastV(+OFT)
 80 [28], SparseVLM [29], and CSP [30]. Parallel efforts emerging as WAMs include Motus [18],
 81 LingBot-VA [31], and Fast-WAM [32]. Despite consistent progress across benchmarks, nearly all
 82 existing action-centric models share a core limitation: treating all action timesteps equally during
 83 training, neglecting the intrinsic physical hierarchy and heterogeneous importance of different motion
 84 phases.

85 2.2 Action Sequence Modeling for Robotics

86 Modeling sequential robotic actions is a core research direction, with early efforts focusing on
 87 trajectory optimization and inverse reinforcement learning (IRL). Recent data-driven approaches
 88 include Action Chunking with Transformers (ACT) [33], which uses transformers to model temporal
 89 dependencies in action sequences, and Diffusion Policy [34], which leverages diffusion models for
 90 smooth, feasible trajectory generation—though these prioritize action quality over critical action
 91 prioritization based on physical characteristics (e.g., velocity). Prior works have explored importance
 92 weighting for imitation learning: some weight entire trajectories by demonstration quality [35, 36],
 93 while others focus on per-timestep weighting (e.g., IRIS [37] for informative offline data, uncertainty-
 94 based weighting for critical states [38]). However, these are limited to single-task learning or require
 95 extra overhead, unlike our velocity-based approach that needs no additional training and is compatible
 96 with arbitrary VLA frameworks. A small number of VLA works explore action weighting (e.g.,
 97 VLA-ADP [21] prunes redundant fast actions for efficiency), but none explicitly link velocity to
 98 learning priority. Different from all prior arts, **AttenA+** introduces a lightweight, plug-and-play
 99 velocity-field weighting principle that emphasizes learning on critical action phases, requiring no
 100 extra supervision and universally compatible with mainstream action-centric foundation models.

101 2.3 Attention Mechanisms in Robotic Learning

102 Attention has become a standard component in modern robotic foundation models, yet its usage
 103 remains largely confined to input modalities. Visual attention focuses on task-relevant spatial regions
 104 for manipulation [39]; language attention aligns linguistic instructions with visual observations
 105 [40]; cross-modal attention further fuses vision and language features to predict actionable policies
 106 [3, 10]. Remarkably, almost all prior attention designs operate exclusively on input vision-language
 107 streams, while output action trajectories are treated as plain unweighted regression targets. Our work
 108 breaks this convention by proposing *action-centric attention*: we apply attention weighting directly
 109 to output action sequences, guided by physical velocity priors to emphasize precision-demanding
 110 motion segments. This extends attention design from input feature alignment to physical-aware action
 111 trajectory modeling, mirroring the hierarchical nature of human motor control.

2.4 Scaling Laws vs. Data Mining in Robotic Learning

The dominant paradigm for improving robotic models relies on scaling laws: expanding dataset size (e.g., RT-1 [6] with 130k trajectories) or model capacity (e.g., PaLM-E [41] with 540B parameters). Inherited from NLP, this is inefficient for heterogeneous robotic action data, where critical and trivial actions coexist. Recent works challenge this paradigm (e.g., DemInf [17] curates high-quality trajectories, physics-guided methods augment data). Our work advances this direction with a *data mining* paradigm: instead of scaling, we mine velocity-based importance patterns in existing sequences to boost performance, offering a cost-effective alternative for data-constrained scenarios.

3 Methodology

3.1 Unified Task Definition and Notation

We unify the formulation for action-centric robotic foundation models (including VLA and WAM) across discriminative and generative paradigms using consistent notation, focusing on the core mapping from multimodal inputs to task-compliant action sequences:

- $\mathcal{I} = \{i_1, i_2, \dots, i_T\}$: Sequence of visual observations (RGB images) at timesteps $1 \leq t \leq T$, where $i_t \in \mathbb{R}^{H \times W \times 3}$ in most cases.
- L : Natural language instruction (e.g., "stack the blue block on the red block"), tokenized to $L = [l_1, l_2, \dots, l_N]$ via standard tokenizers (e.g., CLIP [42], Bert [43]).
- $\mathcal{A} = \{a_1, a_2, \dots, a_T\}$: Robotic action sequence, with $a_t \in \mathbb{R}^D$ and D denoting action dimensions (e.g., $D = 7$ for Libero benchmarks).
- $\mathcal{A}^{gt}$: Ground-truth action sequence from expert demonstrations, serving as the target for model learning.

3.2 Discriminative Action-Centric Models

Discriminative paradigms [3, 19, 10] cast robotic action learning as a deterministic regression task. Given visual observations $\mathcal{I}$ and language instructions L , a discriminative model f_θ (parameters θ) directly predicts a deterministic action sequence:

$$\mathcal{A}^{\text{pred}} = f_\theta(\mathcal{I}, L) \quad (1)$$

Training minimizes the discrepancy between predicted and ground-truth actions via a regression loss over the dataset $\mathcal{D}$ (tuples of $\mathcal{I}, L, \mathcal{A}^{gt}$). Most existing works adopt an unweighted ℓ_1 loss as the optimization objective:

$$\theta^* = \arg \min_{\theta} \mathbb{E}_{(\mathcal{I}, L, \mathcal{A}^{gt}) \sim \mathcal{D}} \left[\frac{1}{T \cdot D} \sum_{t=1}^T \sum_{d=1}^D \left| a_{t,d}^{\text{pred}} - a_{t,d}^{\text{gt}} \right| \right] \quad (2)$$

3.3 Generative Action-Centric Models via Flow Matching

The π -series models (π_0 [5], $\pi_{0.5}$ [12]) are representative generative action-centric frameworks based on flow matching. Instead of direct regression, these models learn a continuous vector field to transform Gaussian random noise $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ into task-aligned action sequences. Specifically, the learnable network g_ϕ predicts a time-dependent flow field conditioned on $\mathcal{I}$ and L , denoising random inputs toward $\mathcal{A}^{gt}$. The standard unweighted flow matching objective is:

$$\phi^* = \arg \min_{\phi} \mathbb{E}_{\substack{(\mathcal{I}, L, \mathcal{A}^{gt}) \sim \mathcal{D} \\ \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})}} \left[\frac{1}{T \cdot D} \sum_{t=1}^T \sum_{d=1}^D \left\| u_t(\epsilon; \mathcal{I}, L) - (a_{t,d}^{\text{gt}} - \epsilon_d) \right\|_2^2 \right] \quad (3)$$

Here, $a_{t,d}^{\text{gt}}$ is the d -th dimension of $\mathcal{A}^{gt}$ at timestep t , with T and D denoting action sequence length and dimensionality, respectively. While $\pi_{0.5}$ introduces hierarchical reasoning for complex tasks, it retains the original flow matching objective. Related diffusion policy frameworks are discussed in Appendix B.

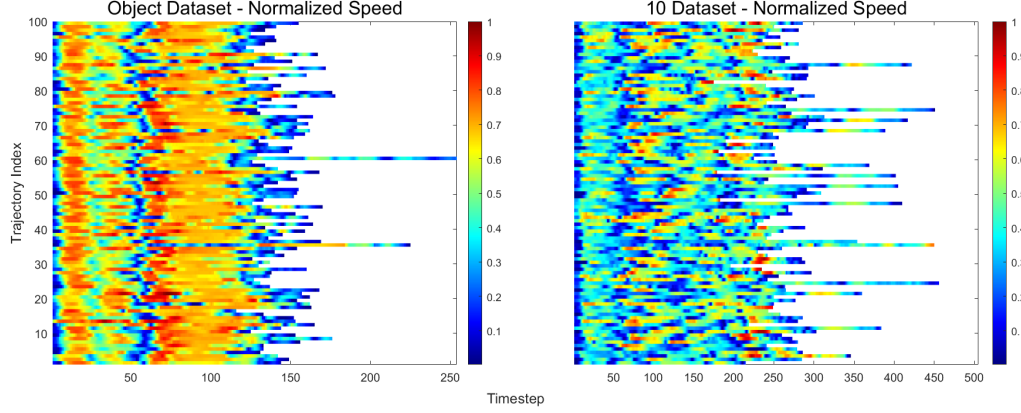


Figure 2: Speed visualization of object/long tasks from the Libero dataset.

150 3.4 Core Limitation Across Paradigms

151 Discriminative and generative action-centric models share a critical limitation: their optimization
 152 objectives assume uniform importance across all action timesteps, conflicting with the physical reality
 153 of robotic manipulation. This limitation is formalized via three key observations:

- 154 1. **Velocity-Criticality Correlation** For any ground-truth action a_t^{gt} , we define its joint velocity
 155 magnitude as:

$$v_t = \|a_t^{gt}\|_2 = \sqrt{\sum_{d=1}^D (a_{t,d}^{gt})^2} \quad (4)$$

156 where $\|\cdot\|_2$ denotes the L_2 norm. In real-world manipulation, slow actions ($\mathcal{A}_S$) correspond
 157 to critical steps (e.g., precise grasping, object placement) where small errors cause failure,
 158 while fast actions ($\mathcal{A}_F$) are transitional and error-tolerant.

- 159 2. **Uniform Objective Misalignment** All baselines assign equal weight to $\mathcal{A}_S$ and $\mathcal{A}_F$ (disjoint
 160 partitions of $\mathcal{A}^{gt}$): For discriminative models: $\mathcal{L}(\mathcal{A}_S^{\text{pred}}, \mathcal{A}_S^{gt}) = \mathcal{L}(\mathcal{A}_F^{\text{pred}}, \mathcal{A}_F^{gt})$ (Equation
 161 2); For generative models: $\mathcal{L}_{\text{FM}}(\mathcal{A}_S) = \mathcal{L}_{\text{FM}}(\mathcal{A}_F)$ (Equation 3). This leads to inefficient
 162 learning: models over-optimize trivial fast actions and under-optimize critical slow actions,
 163 limiting success rates (Table 1).

- 164 3. **Generative Framework Shortcoming** Even flow matching/diffusion-based methods (e.g.,
 165 π_0) that model action feasibility fail to address this—their focus on "smooth action genera-
 166 tion" does not distinguish between critical and non-critical smoothness, degrading perfor-
 167 mance on high-precision tasks.

168 3.5 Velocity-Field Attention (AttenA+)

169 To address uniform weighting, we propose a velocity-aware mechanism compatible with all action-
 170 centric paradigms. Using our benchmark datasets (Libero [1], RoboTwin [2]), we construct a weight
 171 matrix $\mathcal{W} \in \mathbb{R}^{T \times 1}$ where $w_t \propto 1/v_t$ (higher weight for slower, critical actions).

172 3.5.1 Action Velocity Calculation

173 For the Libero dataset ($D = 7$: 6 joint velocities + 1 gripper state), we compute velocity magnitude
 174 using the first 6 dimensions of $\mathcal{A}^{gt}$ (ignoring gripper state) via Equation 4. As shown in Figure 2,
 175 low-speed segments align with critical task phases, while high-speed segments are transitional.

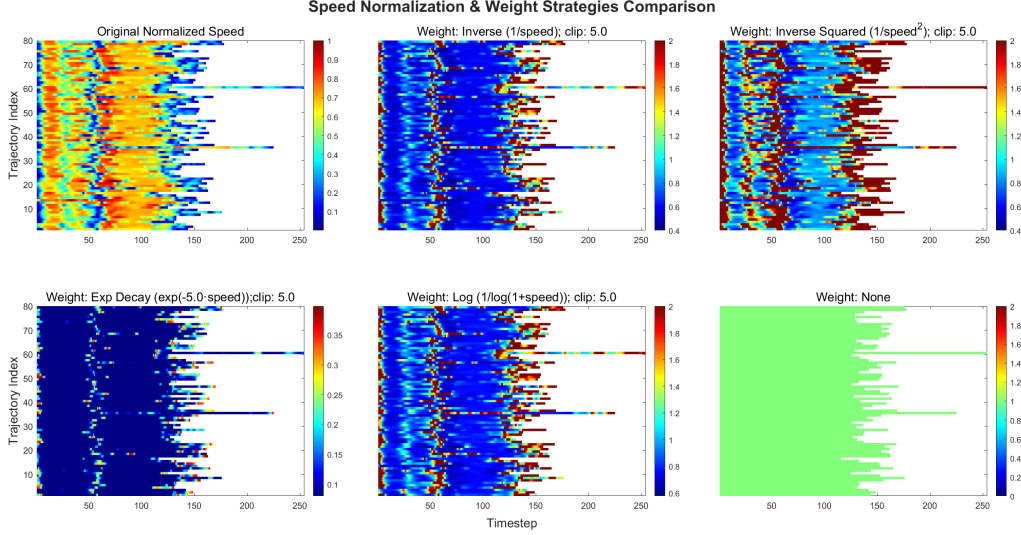


Figure 3: Visualization of the object task dataset with $clip_{\max} = 5.0$.

3.5.2 Velocity-Based Action Attention Weighting Strategies

We design four configurable weighting strategies (inverse, inverse squared, exp decay, log) to map low velocity to high weight $w_{b,t}$, detailed in Appendix C. To stabilize training and avoid extreme weights, we apply two regularization steps:

- **Weight clipping:** Limit weights to $[1/clip_{\max}, clip_{\max}]$;
- **Optional normalization:** Normalize weights to an average of 1 for consistent scaling.

Figure 3 visualizes attention weights from different strategies after regularization ($clip_{\max} = 5.0$), clearly amplifying critical low-speed actions. Additional visualizations and feature analyses across clipping thresholds are in Appendix D.

3.5.3 Revised Optimization Objective for AttenA+

Discriminative Models (AttenA+Disc): The weighted ℓ_1 objective for discriminative models is:

$$\theta^* = \arg \min_{\theta} \mathbb{E}_{(\mathcal{I}, L, \mathcal{A}^{gt}) \sim \mathcal{D}} \left[\frac{1}{T \cdot D} \sum_{t=1}^T \sum_{d=1}^D w_t \cdot |a_{t,d}^{\text{pred}} - a_{t,d}^{gt}| \right] \quad (5)$$

Flow Matching Models (AttenA+FM): The weighted flow matching objective for generative models ($\pi_0/\pi_{0.5}$) is:

$$\phi^* = \arg \min_{\phi} \mathbb{E}_{(\mathcal{I}, L, \mathcal{A}^{gt}) \sim \mathcal{D}} \left[\frac{1}{T \cdot D} \sum_{t=1}^T \sum_{d=1}^D w_t \cdot \left\| u_t(\epsilon; \mathcal{I}, L) - (a_{t,d}^{gt} - \epsilon_d) \right\|_2^2 \right] \quad (6)$$

AttenA+ offers two key advantages: (1) **Paradigm Agnosticism:** Universal compatibility with discriminative and generative action-centric models; (2) **Physical Alignment:** Aligns learning with real-world robotic manipulation physics.

4 Experiment

We evaluate AttenA+ using four metrics: (1) **Success Rate (SR) (%)**: percentage of successfully completed tasks. (2) **Average Success Rate ($\overline{\text{SR}}$) (%)**: mean success rate across tasks. (3) **Average Error Rate ($\overline{\text{ER}}$) (%)**: mean error rate across tasks. (4) **Average Success Rate Improvement ($\overline{\text{SR-I}}$)**

Table 1: Performance on Libero Compared with SOTA Methods. $\overline{\text{SR}}(\%)$: Average Success Rate; $\overline{\text{ER}}(\%)$: Average Error Rate; $\text{SR-I}(\%)$: Average Success Rate Improvement; $\text{RER-R}(\%)$: Average Relative Error Rate Reduction (Compared with AttenA+OFT using Eq. 7).

Method	Spatial	Object	Goal	10	SR $\uparrow$	ER $\downarrow$	SR-I	RER-R
OpenVLA [3]	84.7	88.4	79.2	53.7	76.5	23.5	+22.15	-94.26
SparseVLM [29]	79.8	67	72.6	39.4	64.7	35.3	+33.95	-96.18
FastV [28]	83.4	84	74.2	51.6	73.3	26.7	+25.35	-94.94
FastV(+OFT) [28]	96.8	81	96.4	73	86.8	13.2	+11.85	-89.77
VLA-Cache [27]	83.8	85.8	76.4	52.8	74.7	25.3	+23.95	-94.66
FlashVLA [26]	84.2	86.4	75.4	51.4	74.35	25.65	+24.3	-94.74
SP-VLA [25]	75.4	85.6	84.4	54.2	74.9	25.1	+23.75	-94.62
WorldVLA [11]	85.6	89	82.6	59	79.05	20.95	+19.6	-93.56
WorldVLA* [11]	87.6	96.2	83.4	60	81.8	18.2	+16.85	-92.58
NORA [24]	85.6	87.8	77	45	73.85	26.15	+24.8	-94.84
NORA-Long [24]	92.2	95.4	89.4	74.6	87.9	12.1	+10.75	-88.84
SmolVLA [23]	93	94	91	77	88.75	11.25	+9.9	-88
CogACT [22]	97.2	98	90.2	88.8	93.55	6.45	+5.1	-79.07
CSP [30]	84.7	82.2	77.1	74.3	79.575	20.425	+19.075	-93.39
π_0 + FAST [20]	96.4	96.8	88.6	60.2	85.5	14.5	+13.15	-90.69
π_0 [5]	96.8	98.8	95.8	85.2	94.15	5.85	+4.5	-76.92
UniVLA [10]	96.5	96.8	95.6	92	95.225	4.775	+3.425	-71.73
VLA-ADP [21]	99	98.2	96.8	91.2	96.3	3.7	+2.35	-63.51
OpenVLA-OFT [19]	97.6	98.4	97.9	94.5	97.1	2.9	+1.55	-53.45
AttenA+OFT	99 $\pm$ 0.16	100 $\pm$ 0.00	98.8 $\pm$ 0.28	96.6 $\pm$ 0.30	98.6	1.4	-	-

(%): absolute gain in average success rate. (5) **Average Relative Error Rate Reduction ($\overline{\text{RER-R}}$)**
 (%): relative error reduction computed by

$$\overline{\text{RER-R}} = \left(1 - \frac{\overline{\text{ER}}_{\text{AttenA+}}}{\overline{\text{ER}}_{\text{other}}}\right) \times 100. \quad (7)$$

4.1 Libero Benchmark

We build AttenA+OFT upon the official OpenVLA-OFT framework, and benchmark our approach against representative state-of-the-art VLA and WAM models across all four task subsets of the Libero dataset. We select the best-performing checkpoint from training, then conduct evaluation across 4 random seeds to report the mean and standard deviation of success rates. Additional training configurations are provided in Appendix E.1. As summarized in Table 1, AttenA+OFT obtains an overall average success rate of 98.6%, surpassing the prior SOTA OpenVLA-OFT by 1.5%. Consistent performance gains are observed across all task categories. In particular, our method achieves a 2.1% improvement on long-horizon manipulation tasks, verifying that our action attention mechanism effectively enhances robustness and precision for complex, extended sequential behaviors.

4.2 RoboTwin Benchmark

We further validate our method on the RoboTwin benchmark, implementing AttenA+WAM based on the Fast-WAM framework. As shown in Table 2, AttenA+WAM achieves a new state-of-the-art average success rate of 93.1%, outperforming the prior best LingBot-VA by 0.9%, without requiring any embodied pre-training. This confirms that our action attention mechanism can effectively boost performance even on larger, more diverse real-world benchmarks.

4.3 Improvement of Different Models with Action Attention

As shown in Table 3, we validate the effectiveness and generality of our velocity-field-based action attention by integrating it into both discriminative and generative action-centric models, respectively. For the discriminative framework, we apply our method to OpenVLA-OFT, a strong baseline that has already achieved high performance on the LIBERO benchmark. After equipping it with action attention, the resulting AttenA+OFT yields clear and consistent gains across all task categories: Spatial (+1.4%), Object (+1.6%), Goal (+0.9%), and Long-horizon tasks (+2.1%). Overall, the average success rate improves by +1.5% (from 97.1% to 98.6%), while the average error rate is reduced by -1.5%. For the generative framework, we adopt π_0 (a flow-matching-based model) as the backbone and construct AttenA+ π_0 with our proposed action attention. Similarly, consistent

Table 2: Performance on RoboTwin 2.0 Compared with SOTA Methods. Refer to Appendix F

for detailed results on all 50 tasks.							
Method	Embodied PT.	Clean	Rand.	SR $\uparrow$	ER $\downarrow$	SR-I	RER-R
π_0 [5]	✓	65.92	58.40	62.2	37.8	+30.9	-81.8
$\pi_{0.5}$ [12]	✓	82.74	76.76	79.75	20.25	+13.4	-65.9
X-VLA [44]	✓	72.9	72.8	72.85	27.15	+20.3	-74.6
Motus [18]	✓	88.66	87.02	87.8	12.2	+5.3	-43.3
LingBot-VA [31]	✓	92.90	91.50	92.2	7.8	+0.9	-11.5
Fast-WAM [32]	✗	91.88	91.78	91.8	8.2	+1.3	-15.5
AttenA+WAM (Ours)	✗	93.3	92.9	93.1	6.9	-	-

improvements are observed across all task types, with an average success rate increase of +1.13% and an average error rate reduction of -1.13%.

These results demonstrate that our velocity-field action attention is *paradigm-agnostic* and can serve as a universal plug-and-play enhancement for both discriminative regression-based models and generative flow-matching models. Notably, the performance gain is especially pronounced on long-horizon tasks, where distinguishing critical actions from transitional movements becomes increasingly important for maintaining successful execution. This further confirms that prioritizing critical actions via velocity-guided weighting effectively boosts the reliability and precision of both discriminative and generative policies.

Table 3: Model Performance Improvement with Velocity-Field-Based Action Attention on LIBERO dataset.

Model		Spatial	Object	Goal	10	SR	ER
Discriminative	OpenVLA-OFT	97.6	98.4	97.9	94.5	97.1	2.9
	AttenA+OFT	99.0	100	98.8	96.6	98.6	1.4
	Improvement	+1.4	+1.6	+0.9	+2.1	+1.5	-1.5
Generative	π_0	96.8	98.8	95.8	85.2	94.15	5.85
	AttenA+ π_0	98.1	99.4	96.9	86.7	95.28	4.72
	Improvement	+1.3	+0.6	+1.1	+1.5	+1.13	-1.13

4.4 Real-World Robot Experiments

As shown in Figure 4, we design 4 kinds of task for validation using the Franka manipulator: (a) Close the open drawer, (b) Put the Green Cube into Green Bowl, (c) Put Object-A into Green Bowl, (d) Put Object-A into XXX and then put Object-B into XXX. For the easy tasks (a) and (b), we collect 50 trajectories for demonstration. For harder tasks (c) and (d), we collect 100 trajectories for demonstration. Notably, during demonstration data collection, we use different speed for different phase: at the beginning, we use the baseline speed for approaching the object for grasping, then we change the speed to be 1/3 of the baseline to fine align and operate the object which indicates critical actions. Then after grasping the object we change the speed to baseline and fastly move to the bowl. When approaching the bowl, the speed is again reduced to 1/3 of the baseline for fine align to the bowl and finally release the object. After collection, we clean the trajectory by removing the no action waiting frames and do action smoothing for efficient training and action attention.

We then finetune and test the task following the OpenVLA-OFT recipe with 2 Nvidia H800 GPUs. In the testing phase, we deploy the model on a RTX-4090 GPU, evaluate each task for 50 times and compute the SR. The results are shown in Table 4. We can see that AttenA+OFT consistently outperforms the baseline OpenVLA-OFT across all real-world tasks, improving the average success rate from 92.5% to 97.0%, with the largest gains on the more complex multi-object and long-horizon tasks, further validating the effectiveness of our method in real-world scenarios.

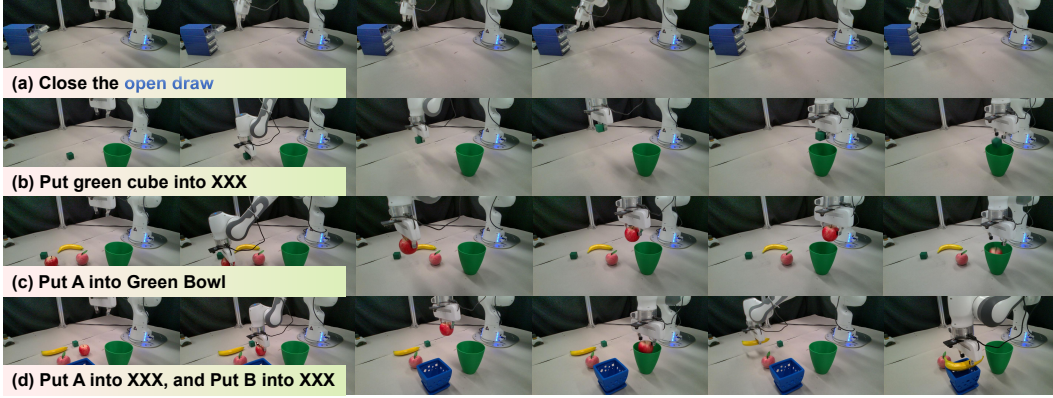


Figure 4: Real Experiments Demonstration on Franka.

Table 4: Real Experiments on Franka. Each test is repeated 50 times and compute the success rate.

Model	Close Draw	Put Cube	Multi-object	Long	SR	ER
OpenVLA-OFT	100	96	90	84	92.5	7.5
AttenA+OFT	100	100	98	90	97.0	3.0
Improvement	0	4	8	6	4.5	-4.5

5 Conclusion

This work presents **AttenA+**, a generic enhancement framework for action-centric robotic foundation models that introduces velocity-field-based action attention to prioritize critical, low-speed manipulation steps during training. By weighting slow, precise actions more heavily in the optimization objective function, AttenA+ aligns model optimization with real-world robotic manipulation physics—without modifying the core architectures of existing discriminative or generative models (including VLA and WAM). Evaluated on standard benchmarks (Libero and RoboTwin), our method consistently improves performance across both paradigms, achieving state-of-the-art results with clear gains in success rate and error reduction. The proposed mechanism is paradigm-agnostic and can be readily extended to other action-centric architectures, including diffusion policies and future robotic foundation models.

Despite these contributions, our work has several limitations that point to promising future directions. First, our current velocity-weighted attention relies on a hand-crafted, speed-based heuristic: we prioritize low-speed actions under the assumption that precision-critical manipulation steps are inherently slow. However, this heuristic does not generalize to all robotic tasks. For dynamic scenarios such as high-speed grasping, table tennis, or reactive collision avoidance, critical actions may instead be fast and ballistic, where our fixed weighting scheme would fail to prioritize these time steps. Second, our attention mechanism only incorporates velocity information, ignoring other physical cues (e.g., force, torque, or jerk) that are equally critical for characterizing action importance in real-world manipulation.

Future work should therefore move beyond fixed heuristic rules to *physically grounded, learnable action attention* that integrates multi-modal physical signals and task-specific constraints, while dynamically adapting to task semantics. By respecting the physical structure of robotic actions rather than treating all timesteps equally, we can build more efficient, robust, and generalizable action-centric robotic foundation systems.

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A Technical Appendices and Supplementary Material

Technical appendices with additional results, figures, graphs and proofs may be submitted with the paper submission before the full submission deadline (see above), or as a separate PDF in the ZIP file below before the supplementary material deadline. There is no page limit for the technical appendices.

B Action Attention Formulation of Generative VLA: Diffusion Policy

B.1 Generative VLA: Diffusion Model (Diffusion Policy)

Diffusion Policy (DP) represents an independent line of generative work, distinct from $\pi_0/\pi_{0.5}$. It models action generation as a iterative denoising process. Given K diffusion steps, the model h_ψ predicts the noise ϵ_k^{pred} added to the action state at step k . The standard unweighted L_2 diffusion optimization objective is:

$$\psi^* = \arg \min_{\psi} \mathbb{E}_{\substack{(\mathcal{I}, L, \mathcal{A}^{gt}) \sim \mathcal{D} \\ k \sim \text{Uniform}(1, K) \\ \epsilon_k \sim \mathcal{N}(0, I)}}} \left[\frac{1}{T \cdot D} \sum_{t=1}^T \sum_{d=1}^D \left\| \epsilon_k^{\text{pred}} - \epsilon_k \right\|_2^2 \right] \quad (8)$$

where $a_{t,d}^{(k)} = \alpha_k a_{t,d}^{gt} + \beta_k \epsilon_k$ is the noisy action, and α_k, β_k are pre-defined diffusion schedule coefficients.

B.2 Revised Optimization Objective for Diffusion Policy (AttenA+Diff)

For diffusion-based generative policies (Diffusion Policy), we apply the same velocity-based weighting to the denoising objective:

$$\psi^* = \arg \min_{\psi} \mathbb{E}_{\substack{(\mathcal{I}, L, \mathcal{A}^{gt}) \sim \mathcal{D} \\ k \sim \text{Uniform}(1, K) \\ \epsilon_k \sim \mathcal{N}(0, I)}}} \left[\frac{1}{T \cdot D} \sum_{t=1}^T \sum_{d=1}^D w_t \cdot \left\| \epsilon_k^{\text{pred}} - \epsilon_k \right\|_2^2 \right] \quad (9)$$

where w_t emphasizes denoising accuracy for slow, critical timesteps during the diffusion process.

C Velocity-Based Action Attention Weighting Strategies

We detail the four handcrafted weighting strategies used to implement velocity-field-based action attention in AttenA+. These formulations serve as *empirical, physics-inspired examples* to demonstrate the core idea of prioritizing slow, critical actions during training, rather than definitive or exclusive solutions. All weighting rules assign higher importance to low-velocity timesteps, consistent with the intuition that precise manipulation phases require stricter optimization in our experiment datasets (Libero, RoboTwin).

For the b -th sample at timestep t , the velocity-aware weight $w_{b,t}$ is defined as follows:

1. Inverse strategy (baseline)

$$w_{b,t} = \frac{1}{v_{b,t}} \quad (10)$$

This baseline scheme applies inverse weighting proportional to action speed, providing a mild but clear emphasis on slower movements.

2. Inverse squared strategy (amplified weight difference)

$$w_{b,t} = \frac{1}{v_{b,t}^2} \quad (11)$$

By squaring the velocity term, this strategy strongly amplifies the contrast between slow and fast actions, making it the default choice in our main experiments.

3. Exponential decay strategy (fast attenuation)

$$w_{b,t} = e^{-\alpha \cdot v_{b,t}} \quad (12)$$

where $\alpha = 5.0$ controls the decay rate. This method suppresses high-speed actions rapidly while maintaining soft weighting for slow segments.

4. Logarithmic strategy (smoothed weight)

$$w_{b,t} = \frac{1}{\log(1 + v_{b,t})} \quad (13)$$

The logarithmic transform yields gentle, stable weighting, reducing sensitivity to noise in velocity estimation.

Notably, these four heuristic functions are *example implementations* chosen for simplicity, interpretability, and empirical effectiveness. They are not intended to limit the design space of action attention. In future work, action weighting can be naturally extended to broader families of parametric functions, task-adaptive formulations, or *fully learnable attention mechanisms* that infer importance end-to-end from data and physical constraints, rather than relying on fixed handcrafted rules.

D Visualization of Action Speed and Velocity-Guided Action Attention

In this section, we provide detailed analysis of the action speed patterns and velocity-guided attention weights, which are briefly summarized in the main text.

Figures 5–8 present comprehensive visualizations of raw action velocity profiles and the resulting velocity-field-based attention weights under four distinct clipping thresholds $clip_{\max} \in \{1.0, 2.0, 5.0, 10.0\}$ on the Libero-object manipulation benchmark. Each figure follows a consistent layout: Subplot 1 illustrates the temporal distribution of raw action speed magnitudes across multiple expert demonstration trajectories, revealing inherent speed variations within task execution. Subplots 2 through 5 display the attention weight distributions generated by the four velocity transformation rules (Equations 10–13): inverse weighting, inverse squared weighting, exponential decay weighting, and logarithmic weighting, respectively. Subplot 6 serves as the baseline, showing the original action trajectory under uniform, unweighted treatment without action attention.

From the raw velocity visualizations, distinct slow–fast motion patterns emerge consistently across all task trajectories. Slow-motion segments consistently align with task-critical phases, including robot initialization, precise object approaching, fine manipulation, grasping, targeted placement, and

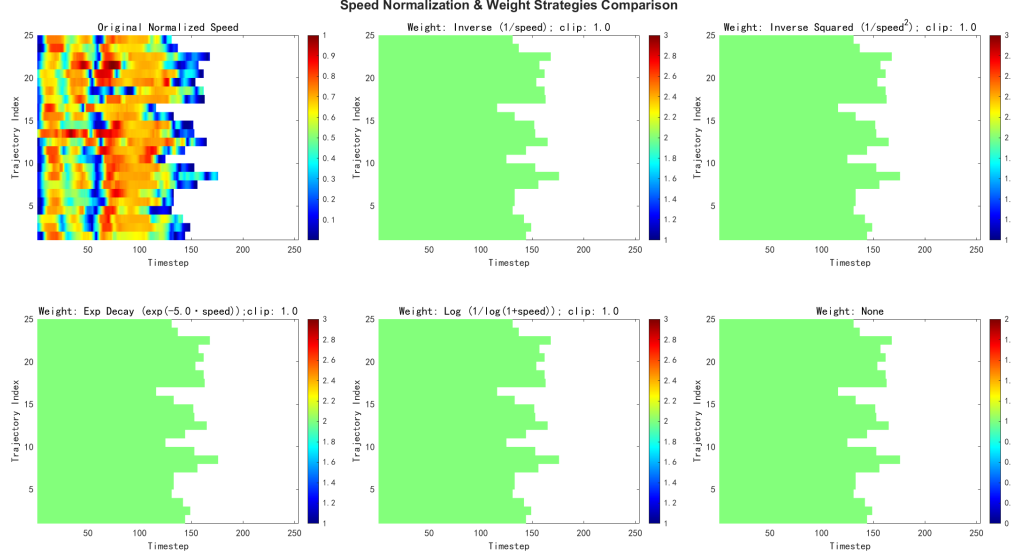


Figure 5: Visualization of Action Speed in Libero-Object Task with Different $clip_{max} = 1.0$

task completion. These phases demand high positional accuracy and are highly sensitive to execution errors, making them decisive for overall task success. Conversely, fast-motion segments correspond to robust, error-tolerant transitional movements, such as free-space arm traversal, coarse positioning toward target objects, and post-grasp repositioning, where minor deviations rarely lead to task failure.

The effect of the clipping threshold $clip_{max}$ is clearly demonstrated across Figures 5–8. At $clip_{max} = 1.0$, all velocity-adaptive weighting schemes collapse to uniform values, reverting AttenA+ to a standard VLA model with equal emphasis on all timesteps. As $clip_{max}$ increases sequentially from 1.0 to 2.0, 5.0, and 10.0, the discriminative power of action attention is progressively strengthened: slow critical actions receive increasingly prominent weights, while fast transitional actions are assigned relatively lower weights, widening the gap in learning priority.

Moreover, the four velocity mapping functions exhibit distinctive attention characteristics. As clearly visible in the $clip_{max} = 2.0$ visualization, exponential decay weighting (Equation 12) produces highly localized emphasis: it strongly amplifies a small set of extremely slow actions while broadly suppressing fast actions across a wide range. In contrast, inverse (Equation 10), inverse squared (Equation 11), and logarithmic (Equation 13) schemes maintain widespread emphasis on slow actions and exert mild, localized suppression on fast actions. Within this group, the intensity of low-speed amplification follows a clear hierarchy: inverse squared (Equation 11) yields the strongest enhancement, followed by logarithmic weighting (Equation 13), and then inverse weighting (Equation 10). This consistent trend is observable across all clipping thresholds in Figures 5–8, validating the design principles of velocity-field-based action attention and supporting the selection of inverse squared weighting as the default configuration in the main experiments.

E Details about Model Training

This section presents comprehensive training and implementation details for AttenA+OFT (evaluated on LIBERO) and AttenA+WAM (evaluated on RoboTwin), covering architectural modifications, optimization configurations, fine-tuning pipelines, checkpoint scheduling, and best-model selection criteria. All experiments in this work use the weight clipping settings $clip_{max} = 2.0$ and $clip_{max} = 5.0$, which are applied consistently across both model variants.

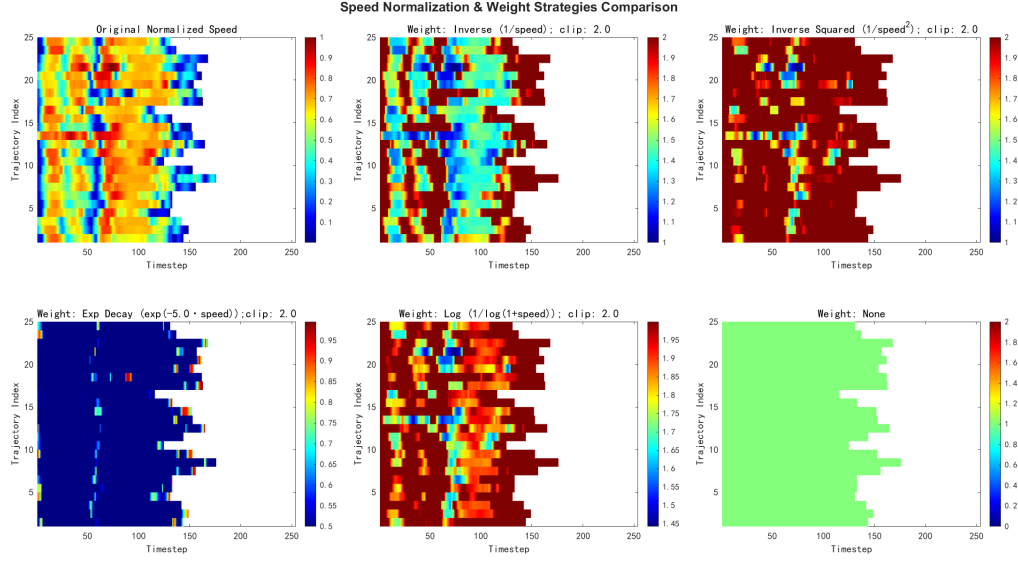


Figure 6: Visualization of Action Speed in Libero-Object Task with Different $clip_{max} = 2.0$

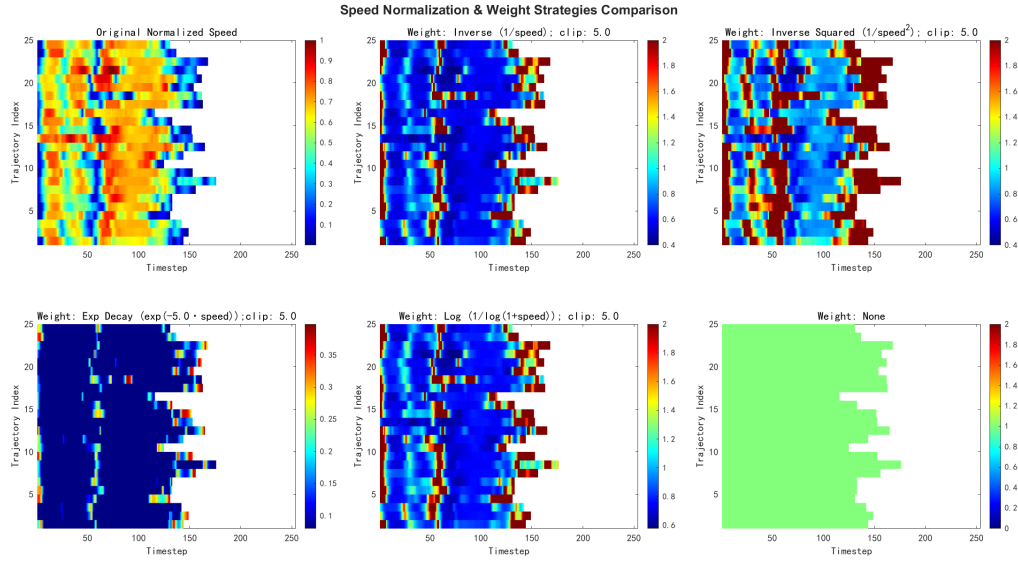


Figure 7: Visualization of Action Speed in Libero-Object Task with Different $clip_{max} = 5.0$

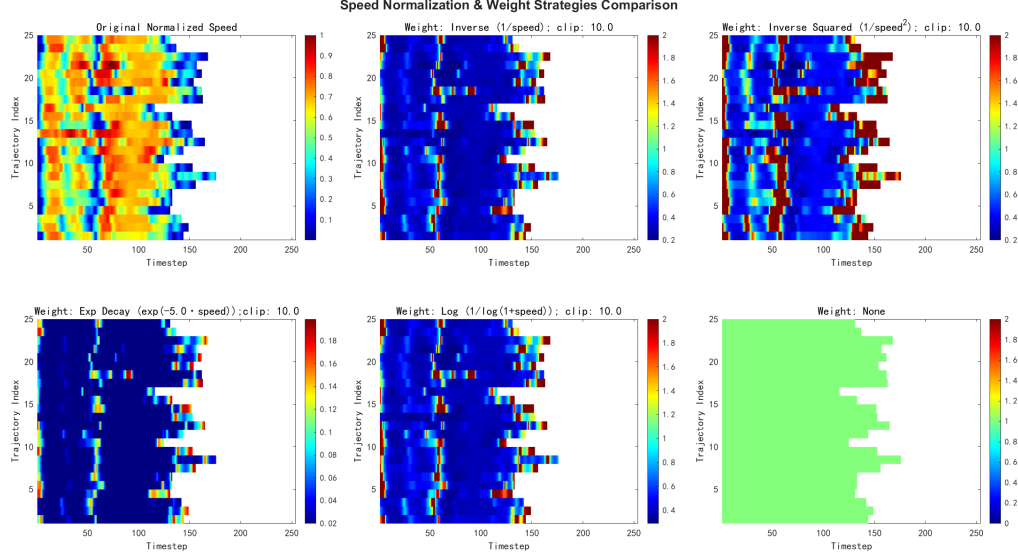


Figure 8: Visualization of Action Speed in Libero-Object Task with Different $clip_{max} = 10.0$

480 E.1 -OFT

481 We build AttenA+OFT as a direct adaptation of the OpenVLA-OFT framework, with our core velocity-
 482 field action attention integrated as a weighted module without altering the backbone architecture.
 483 Following the standard OpenVLA-OFT fine-tuning protocol, we train separate models for each of the
 484 four LIBERO task categories (Spatial, Object, Goal, Long) to ensure fair comparison with prior work.
 485 All models are trained for a total of 200,000 steps, with checkpoints saved every 5,000 steps. The
 486 training time of each model using single H800 GPU is about 35 hours. During training, we strictly
 487 retain the original optimizer configuration, learning rate schedule, batch size, and data preprocessing
 488 used in OpenVLA-OFT to isolate the improvement brought by action attention. After training, we
 489 evaluate over saved checkpoints on the corresponding LIBERO test split and select the checkpoint
 490 with the highest success rate as the final model for reporting results.

491 E.2 AttenA+WAM

492 We implement AttenA+WAM on top of the Fast-WAM architecture, again integrating our velocity-
 493 guided action attention as a plug-and-play weighting module. Since the official Fast-WAM repository
 494 does not release end-to-end fine-tuning code, we adopt a practical and fair adaptation protocol: we
 495 freeze all vision encoders and the pre-trained WAM backbone, and only fine-tune the final action
 496 head using our proposed action attention mechanism. This design ensures we only introduce our
 497 method while preserving the pre-trained knowledge of the original model. We fine-tune on the
 498 RoboTwin dataset for 1 epoch, saving checkpoints every 2,000 steps. The training time of the model
 499 using two H800 GPUs is about 4 days. Consistent with AttenA+OFT, we evaluate over intermediate
 500 checkpoints and select the best-performing one based on validation success rate for final experimental
 501 comparisons.

502 F Detailed Evaluation Results on RoboTwin 2.0 and Real Franka Robot 503 Experiments

Table 5: We present quantitative results on the RoboTwin 2.0 simulation benchmark, covering 50 bimanual manipulation tasks with two difficulty levels. RoboTwin 2.0 serves as a rigorous dual-arm manipulation testbed that demands precise bilateral coordination. The easy setting adopts fixed initial scene arrangements, whereas the hard setting introduces randomized object placements and scene configurations for higher generalization challenges.

Simulation Task	Horizon	Ours		LingBot-VA [31]		π_0 [5]		$\pi_{0.5}$ [12]		X-VLA [44]		Motus [18]	
		Easy	Hard	Easy	Hard	Easy	Hard	Easy	Hard	Easy	Hard	Easy	Hard
Adjust Bottle	1			90%	94%	99%	95%	100%	99%	100%	100%	89%	93%
Beat Block Hammer	1			96%	98%	79%	84%	96%	93%	92%	88%	95%	88%
Blocks Ranking RGB	3			99%	98%	80%	63%	92%	85%	83%	83%	99%	97%
Blocks Ranking Size	3			94%	96%	14%	5%	49%	26%	67%	74%	75%	63%
Click Alarmclock	1			99%	100%	77%	68%	98%	89%	99%	99%	100%	100%
Click Bell	1			100%	100%	71%	48%	99%	66%	100%	100%	100%	100%
Dump Bin Big Binbin	1			89%	96%	88%	83%	92%	97%	79%	77%	95%	91%
Grab Roller	1			100%	100%	98%	94%	100%	100%	100%	100%	86%	100%
Handover Block	2			99%	78%	47%	31%	66%	57%	73%	37%	86%	73%
Handover Mic	2			94%	96%	97%	97%	98%	97%	0%	0%	78%	63%
Hanging Mug	2			40%	28%	14%	11%	18%	17%	23%	27%	38%	38%
Lift Pot	1			100%	99%	80%	72%	96%	85%	99%	100%	96%	99%
Move Can Pot	1			94%	97%	68%	48%	51%	55%	89%	86%	34%	74%
Move Pillowbottle Pad	1			99%	99%	67%	46%	84%	61%	73%	71%	93%	96%
Move Playingcard Away	1			100%	99%	74%	65%	96%	84%	93%	98%	100%	96%
Move Stapler Pad	1			91%	79%	41%	24%	56%	42%	78%	73%	83%	85%
Open Laptop	1			92%	94%	71%	81%	90%	96%	93%	100%	95%	91%
Open Microwave	1			82%	86%	4%	32%	34%	77%	79%	71%	95%	91%
Pick Diverse Bottles	2			89%	82%	69%	31%	81%	71%	58%	36%	90%	91%
Pick Dual Bottles	2			100%	99%	59%	37%	93%	63%	47%	36%	96%	90%
Place A2B Left	1			97%	93%	43%	47%	87%	82%	48%	46%	82%	79%
Place A2B Right	1			97%	95%	39%	84%	87%	84%	36%	39%	90%	87%
Place Bread Basket	2			97%	95%	62%	46%	77%	64%	81%	71%	91%	94%
Place Bread Skillet	2			95%	90%	66%	49%	85%	66%	77%	67%	86%	83%
Place Burger Fries	2			97%	95%	81%	76%	94%	87%	94%	94%	98%	98%
Place Can Basket	2			81%	84%	55%	46%	62%	62%	49%	52%	81%	76%
Place Cans Plasticbox	2			100%	99%	63%	45%	94%	84%	97%	98%	98%	94%
Place Container Box	1			99%	97%	97%	92%	99%	95%	97%	95%	98%	99%
Place Dual Shoes	2			94%	89%	59%	51%	75%	75%	79%	88%	93%	87%
Place Empty Cup	1			100%	100%	91%	85%	100%	99%	100%	98%	99%	98%
Place Fan	1			99%	93%	66%	71%	87%	85%	80%	75%	91%	87%
Place Mouse Pad	1			93%	96%	20%	20%	60%	39%	70%	70%	66%	68%
Place Object Basket	2			91%	88%	67%	70%	80%	76%	44%	39%	81%	87%
Place Object Scale	1			96%	95%	57%	52%	86%	80%	52%	74%	88%	85%
Place Object Stand	1			99%	96%	82%	68%	91%	85%	86%	88%	98%	97%
Place Phone Stand	1			97%	97%	49%	53%	81%	81%	88%	87%	87%	86%
Place Shoe	1			98%	98%	76%	76%	92%	93%	96%	95%	99%	97%
Press Stapler	1			85%	82%	44%	37%	87%	83%	92%	98%	93%	98%
Put Bottles Dustbin	3			87%	91%	65%	56%	84%	79%	77%	81%	81%	79%
Put Object Cabinet	2			85%	87%	73%	60%	80%	79%	46%	48%	88%	71%
Rotate QRcode	2			96%	91%	74%	70%	89%	87%	34%	33%	89%	73%
Scan Object	2			96%	91%	55%	42%	72%	65%	14%	36%	67%	66%
Shake Bottle Horizontally	1			100%	99%	98%	92%	99%	99%	100%	100%	100%	98%
Shake Bottle	1			100%	97%	94%	91%	99%	97%	99%	100%	100%	97%
Stack Blocks Three	3			99%	98%	72%	52%	91%	76%	6%	10%	91%	95%
Stack Blocks Two	2			100%	98%	93%	79%	97%	100%	92%	87%	100%	98%
Stack Bowls Three	3			86%	83%	77%	75%	77%	71%	76%	86%	79%	87%
Stack Bowls Two	2			94%	98%	94%	95%	95%	96%	96%	93%	98%	98%
Stamp Seal	1			96%	97%	46%	33%	79%	55%	76%	82%	93%	92%
Turn Switch	1			44%	45%	41%	42%	62%	54%	40%	61%	84%	78%
Average (%)	—			92.93	91.55	65.92	58.40	82.74	76.76	72.80	72.84	88.66	87.02

504 NeurIPS Paper Checklist

505 The checklist is designed to encourage best practices for responsible machine learning research,
506 addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove
507 the checklist: **The papers not including the checklist will be desk rejected.** The checklist should
508 follow the references and follow the (optional) supplemental material. The checklist does NOT count
509 towards the page limit.

510 Please read the checklist guidelines carefully for information on how to answer these questions. For
511 each question in the checklist:

- 512 • You should answer [Yes], [No], or [NA].
- 513 • [NA] means either that the question is Not Applicable for that particular paper or the
514 relevant information is Not Available.
- 515 • Please provide a short (1–2 sentence) justification right after your answer (even for NA).

Table 6: Detailed Evaluation Results on Real Franka Robot Experiments. We report the success rates (SR) across four manipulation tasks under the Baseline and our Attention-based method.

Experiment	Baseline				AttenA+			
	Close Draw	Put Cube	Multi-object	Long	Close Draw	Put Cube	Multi-object	Long
1	1	1	1	1	1	1	1	1
2	1	1	1	0	1	1	1	1
3	1	1	1	1	1	1	1	1
4	1	1	1	1	1	1	1	1
5	1	1	1	1	1	1	1	1
6	1	1	0	1	1	1	1	1
7	1	1	1	0	1	1	1	1
8	1	1	1	1	1	1	1	1
9	1	1	1	1	1	1	1	0
10	1	1	1	1	1	1	1	1
11	1	1	1	1	1	1	1	1
12	1	0	1	0	1	1	1	1
13	1	1	1	0	1	1	1	1
14	1	1	0	1	1	1	1	1
15	1	1	1	1	1	1	1	1
16	1	1	0	1	1	1	1	1
17	1	1	1	1	1	1	1	1
18	1	1	1	0	1	1	1	1
19	1	1	1	1	1	1	1	1
20	1	1	1	1	1	1	1	1
21	1	1	1	1	1	1	1	1
22	1	1	1	1	1	1	1	0
23	1	1	1	1	1	1	1	1
24	1	1	1	1	1	1	1	1
25	1	1	1	1	1	1	1	1
26	1	1	1	1	1	1	1	1
27	1	1	1	1	1	1	1	1
28	1	1	1	1	1	1	1	1
29	1	1	1	1	1	1	0	1
30	1	1	1	1	1	1	1	0
31	1	1	1	1	1	1	1	1
32	1	1	1	1	1	1	1	1
33	1	1	1	1	1	1	1	1
34	1	1	0	1	1	1	1	1
35	1	1	1	1	1	1	1	1
36	1	1	1	1	1	1	1	1
37	1	1	1	1	1	1	1	1
38	1	0	1	1	1	1	1	1
39	1	1	1	0	1	1	1	1
40	1	1	1	1	1	1	1	1
41	1	1	1	1	1	1	1	0
42	1	1	1	1	1	1	1	1
43	1	1	0	0	1	1	1	1
44	1	1	1	1	1	1	1	0
45	1	1	1	1	1	1	1	1
46	1	1	1	1	1	1	1	1
47	1	1	1	0	1	1	1	1
48	1	1	1	1	1	1	1	1
49	1	1	1	1	1	1	1	1
50	1	1	1	1	1	1	1	1
SR (%)	100	96	90	84	100	100	98	90

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will be asked to also include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While "[Yes]" is generally preferable to "[No]", it is perfectly acceptable to answer "[No]" provided a proper justification is given (e.g., "error bars are not reported because it would be too computationally expensive" or "we were unable to find the license for the dataset we used"). In general, answering